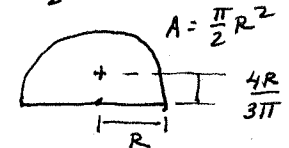
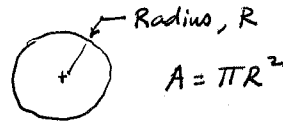
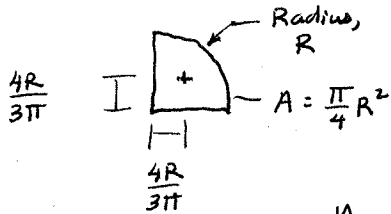
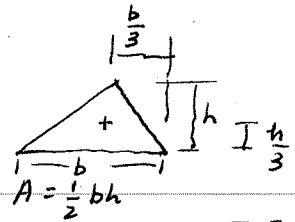
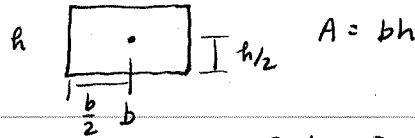


STATICS OCI Problem :

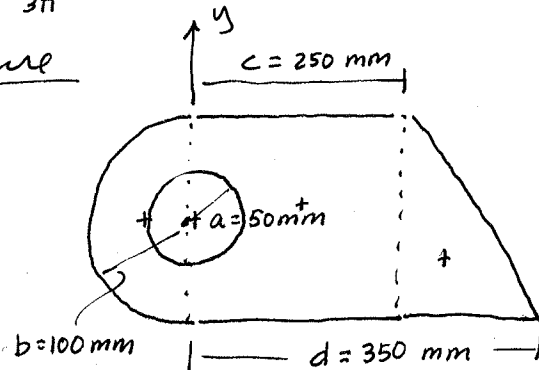
CENTROID OF A COMPOSITE AREA

[NOTE: A "composite" area is an area made up of multiple sub-areas for which we know their areas and centroids.]

Typical Sub-Areas :



Procedure



NOTES:

$$\bar{x} = \frac{\sum \tilde{x}_i A_i}{\sum A_i}; \quad \bar{y} = \frac{\sum \tilde{y}_i A_i}{\sum A_i}$$

$\tilde{x}_i, \tilde{y}_i$ can be (+) or (-) if in $\pm$ x or y direction from axes.

A_i , Area is NEGATIVE if subtracted.... (e.g. a hole....)

SEGMENT NO.	PICTURE	A_i	$\tilde{x}_i$	$\tilde{y}_i$	$\tilde{x}_i A_i$	$\tilde{y}_i A_i$
①		$\frac{\pi}{2} (100)^2 = +15,708$	Neg! $-\frac{4(100)}{3\pi} = -42.44$	100	$-666.65 \cdot 10^3$	$1570.8 \cdot 10^3$
②		Hole! Neg! $-\pi (50)^2 = -7854$	Zero! 0	100	0	$-785.4 \cdot 10^3$
③		$250(200) = 50,000$	125	100	$+6250 \cdot 10^3$	$+5000 \cdot 10^3$
④		$\frac{1}{2}bh = \frac{1}{2}(100)(200) = 10,000$	$250 + \frac{100}{3} = 283.3$	$\frac{200}{3} = 66.7$	$+2833 \cdot 10^3$	$666.67 \cdot 10^3$

FINALLY,

$$\bar{x} = \frac{\sum \tilde{x}_i A_i}{\sum A_i} = \frac{8416.7 \cdot 10^3 \text{ mm}^3}{67,854 \text{ mm}^2}$$

$$\bar{x} = 124.0 \text{ mm}$$

$$\bar{y} = \frac{\sum \tilde{y}_i A_i}{\sum A_i}$$

$$= \frac{6452.1 \cdot 10^3}{67,854}$$

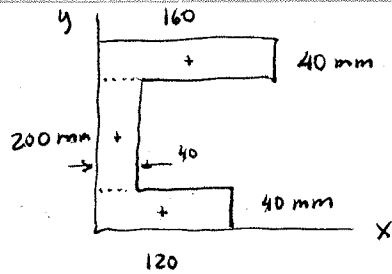
$$\bar{y} = 95.09 \text{ mm}$$

$$\bar{x} = 124.04 \text{ mm}$$

$$\sum A_i = 67,854 \text{ mm}^2$$

$$\sum \tilde{x}_i A_i = 8416.7 \cdot 10^3 \text{ mm}^3 = \sum \tilde{y}_i A_i = 6452.1 \cdot 10^3 \text{ mm}^3$$

7-39



$$\bar{X} = \frac{\sum \tilde{x}_i A_i}{\sum A_i}$$

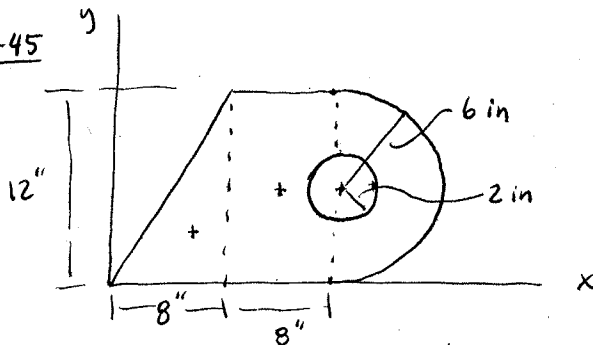
$$= \frac{[60(120)(40) + 20(120)(40) + 80(40)(160)]}{[120(40) + 120(40) + (40)(160)]}$$

$$= \frac{896,000}{16,000} \quad \boxed{\bar{X} = 56 \text{ mm}}$$

$$\bar{y} = \frac{\sum \tilde{y}_i A_i}{\sum A_i} = \frac{[20(40)(120) + 100(40)(120) + 180(40)(160)]}{16,000}$$

$$\bar{y} = \frac{1,728,000}{16,000} = \boxed{108 \text{ mm} = \bar{y}}$$

7-45



$$A = \sum A_i$$

$$= \frac{1}{2}(8)(12) + 8(12) + \frac{\pi}{2}(6)^2 - \pi(2)^2$$

$$= 187.98 \text{ in}^2$$

$$\bar{X} = \frac{\sum \tilde{x}_i A_i}{\sum A_i}$$

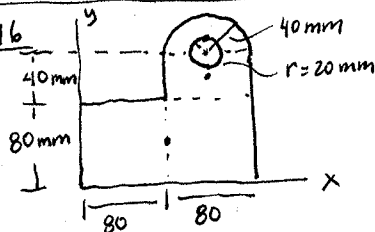
$$\bar{X} = \frac{(\frac{2}{3}8)(\frac{8}{2})(12) + 12(8)(12) + (16 + \frac{4(6)}{3\pi})(\frac{\pi}{2}6^2) - (16)(\pi \cdot 2^2)}{187.98 \text{ in}^2}$$

$$\bar{X} = \frac{2255.7}{187.98} = \boxed{12.0 \text{ in} = \bar{X}}$$

$$\bar{y} = \frac{\sum \tilde{y}_i A_i}{A_T} = \frac{(\frac{12}{3})(\frac{8}{2})(12) + 6(8)(12) + 6\frac{\pi}{2}6^2 - 6 \cdot \pi(2)^2}{187.98}$$

$$\bar{y} = \frac{1031.9}{187.98} = \boxed{5.49 \text{ in} = \bar{y}}$$

7-46



$$A = \sum A_i = 80(160) + 40(80) + \frac{\pi}{2}(40)^2 - \pi(20)^2$$

$$= 17,256.6 \text{ mm}^2$$

$$\bar{X} = \frac{\sum \tilde{x}_i A_i}{A} = \frac{80(80)(160) + 120(40)(80) + 120(\frac{\pi}{2}40^2) - 120(\pi(20)^2)}{17,256.6}$$

$$= \frac{1,558,800}{17,256.6} = \boxed{90.33 \text{ mm} = \bar{X}}$$

$$\bar{y} = \frac{\sum \tilde{y}_i A_i}{A} = \frac{40(80)(160) + 100(40)(80) + (120 + \frac{4(40)}{3\pi})(\frac{\pi}{2}40^2) - 120(\pi(20)^2)}{17,256.6}$$

$$= \frac{1,025 \cdot 10^6}{17,256.6} = \boxed{59.42 \text{ mm} = \bar{y}}$$